

# 8257 Direct Memory Access Controller (DMAC) MPCA Module 4

> **Definition:** The **Intel 8257** is a 4-channel programmable **Direct Memory Access Controller (DMAC)** that enables high-speed data transfer directly between I/O peripherals and system RAM without CPU intervention, bypassing the slow CPU fetch-execute cycle.

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## 1. Detailed Technical Explanation

### 1. The DMA Concept & Speed Advantage

In standard programmed I/O, every byte transferred from a disk/network card to RAM requires multiple CPU instructions (`IN AL, DX` followed by `MOV [SI], AL`). DMA bypasses the CPU:

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## 2. 8257 Internal Block Diagram & Channels

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## 3. DMA Transfer Sequence (Step-by-Step)

- Peripheral asserts **DMA Request** (`DRQ`) to 8257.
- 8257

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## 4. DMA Transfer Modes

1. **Single Transfer Mode:** 8257 transfers one byte per request, releasing the bus back to CPU between transfers.

2. **Block Transfer Mode:**

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## 5. Must-Write Points for Exams

- 8257 has **4 independent DMA channels** (Channel 0 to Channel 3).

- Each channel has its own **16-bit address counter** and **16-bit data counter**.

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## 6. Quick Recall Flow

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DRQ - Data Request